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Sampling of the cowpea curculio, *Chalcodermus aeneus*, with traps in southern peas

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ABSTRACT

The cowpea curculio, *Chalcodermus aeneus* (Boheman) (Coleoptera: Curculionidae), is the key pest of southern pea or cowpea, *Vigna unguiculata* (L.), in Georgia and elsewhere in the southeastern USA. There has never been an effective trapping method reported for this pest. We developed a modified Tedder's trap that was tested against standard boll weevil traps and yellow sticky traps. The new trap detected *C. aeneus* adults earlier and in greater numbers than the other trapping methods. The new trap was used to monitor weekly movement of *C. aeneus* adults for two annual cycles in 2012 and 2013 at multiple locations. The detection of adults was consistent with early spring movement from overwintering sites followed by a summer generation and a fall generation. No adults were collected in traps from the end of December to the end of March. The temporal distribution of *C. aeneus* in traps over the season suggests three distinct periods of adult activity from mid-April to mid-June, late-July to early-October and then mid-November to mid-December. The utility of monitoring *C. aeneus* for regional management strategies is discussed.

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1. Introduction

The cowpea curculio, *Chalcodermus aeneus* (Boheman) (Coleoptera: Curculionidae), is the key pest of southern peas or cowpea, *Vigna unguiculata* (L.) in the southeastern USA (Chalfant 1985). One of the earliest mentions of weevils attacking cowpea in the USA was in Kansas with reference to the cowpea weevil (TenEyck and Call 1909). However, it is not clear if this was *Callosobruchus maculatus* (Fabricius) if the currently approved ESA common name was used correctly, or *C. aeneus*. Ainslie (1910) clearly places the distribution of *C. aeneus* in the USA in 1910 in eastern Texas, Louisiana, Alabama, Florida, Georgia, South Carolina, North Carolina, and less frequently in Oklahoma, Maryland, and Missouri, stating that, "the distribution of this weevil probably coincides with that of the cowpea". The one clear exception to this statement was the lack of *C. aeneus* in California where cowpeas were grown by 1920 (Morse 1920) yet this pest has never been reported as a problem in California (Hall and Frate 1996). More recently, the distribution of *C. aeneus* has been summarized for Latin America: México, Belize, Guatemala, Honduras, and Nicaragua (Jean-Michel and O'Brien 1990), and for *Chalcodermus* sp. in Brazil and Jamaica (Daoust et al. 1985). One thing is clear from these reports; where it occurs, *C. aeneus* has always been a key pest of southern peas.

The cowpea curculio is a small (4.8–5.5 mm in length), shiny, dark weevil with deeply pitted elytra and punctate thorax. The life cycle for this insect is approximately 30–40 days from egg to adult (Capinera 2001). Adults feed on the pods and oviposit on pods or directly into the seed through the pod wall. The egg hatches within the pod and the larva goes through four developmental instars as it feeds on the peas causing severe damage. The 4th instar reaches a length of 7 mm and then chews out of the pod, falling to the ground, and burrowing 2.5–7.6 cm into the soil where it creates a pupal cell in the soil and pupates. The duration of the pupal stage is 5 to 19 days, at which time the non-overwintering adults emerge from the soil, mate, and begin laying eggs in the same or subsequent southern pea plantings (Arant 1938). If green peas are harvested before the larva exits the pod, the grubs become a contaminant in mechanically harvested peas, which causes further quality loss besides the direct loss of marketable peas. The main management tactic for the control of this pest is the use of insecticides applied twice weekly during the pod development stage of the crop. Destruction of alternate or overwintering hosts or crop residue has been the main cultural control recommended and no effective commercial biological controls have been demonstrated (Capinera 2001). In Georgia, both destruction of crop residue and rotation away from heavily infested sites have been emphasized, but the efficacy of these recommendations has not been quantified.

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Vigna unguiculata is a drought-tolerant (Ewansiha and Singh 2006), economically important (Coolong et al. 2012), and highly nutritious crop (Ng and Marechal 1985). In Georgia, southern pea acreage was 57,465 ha in 1951 with a price of $4/bushel (Anon, 1957). This acreage declined to a low of 1,745 ha in 1997 despite higher prices (Adams and Riley 1997). Cowpeas are produced primarily as a food crop and the inability to consistently control the cowpea curculio is one of the main reasons for the decline in fresh frozen pea production. There is a near zero tolerance for curculio-infested peas in the frozen pack process (C.T. Harvey, Fresh Frozen Inc., personal communication). The number of cowpea hectares in Georgia have increased from 2,146 in 2007, to 2,779 in 2008, and 3,299 in 2009 (Anon, 2010). Prices paid to growers for cowpeas have increased from ~$0.19/0.454 kg (lb) of shelled fresh southern peas in 1975 to $0.30 in 2008 and then to $0.50 in 2014 (C.T. Harvey, Fresh Frozen Inc., personal communication). The farm gate value of the crop in 2009 in Georgia was over $12 million (Anon, 2010). With the increased acreage and demand for southern peas, have come increased concerns of *C. aeneus* damage. In 2010 and 2011, multiple consultant reports of *C. aeneus* outbreaks and field control failures were the incentive to conduct new field tests on this key pest. Based on our recent field observations and previous studies (Chalfant 1996), *C. aeneus* tolerance to pyrethroid insecticides (N'Guessan and Chalfant 1990) is a potential problem in Georgia. Pyrethroids are currently the only effective labeled insecticides used against *C. aeneus* in Georgia (Sparks and Riley 2012).

The lack of a sampling method for *C. aeneus* that can detect movement of adults or that can quantify movement in the cowpea landscape has been lacking since the beginning of cowpea cultivation in the USA. We currently know that *C. aeneus* overwinters on various weed hosts such as the following: narrow-leaved vetch, *Vicia sativa* ssp. *nigra* (L.), purple cudweed, *Gnaphalium pupureum* L., heartwing sorrel, *Rumex hastatulus* L., cutleaf evening primrose *Oenothera laciniata* L., and moss verbena *Verbena tenuisecta* Briquet, as early as mid-April and seems to move into cowpeas by mid-May based on sampling date collected with sweep net and direct visual sampling of plants (Sudbrink et al. 1998). The potential for early outbreaks of *C. aeneus* could be assessed if an effective trapping method for adult movement could be developed, as has been done with other curculio species (Tedders and Wood 1994, Braman et al. 2003). In May and June, *C. aeneus* can reproduce on snapbean pods, *Phaseolus vulgaris* L., before southern pea plants bloom so nearby plantings of snap bean can exacerbate the problem. Later during the southern pea growing season *C. aeneus* can be found on sicklepod, *Senna obtusifolia* (L.). Cowpea curculio can also produce an overwintering generation on the beans of *Strophostyles* plants (Sudbrink et al. 1998). An effective trapping method could provide more efficient quantitative evidence of the movement and development of curculio populations in the landscape.

The objectives of this study were to develop an effective trapping method for *C. aeneus* adults and to use this method to detect possible adult movement. Boll weevil trapping and eradication programs indicate that *C. aeneus* can be collected in boll weevil traps (Jones and Williams 2001). Previous pepper weevil (*Anthonomus eugenii* Cano) trapping studies suggest that weevils can be captured on unbaited sticky traps (Riley and Schuster 1993). Thus, we compared standard boll weevil or Leggett traps (Leggett and Cross 1971) and a modified Tedders trap (Tedders and Wood 1994) to a simple, unbaited yellow sticky trap. The null hypothesis was that there would be no difference between traps. We then used the modified Tedders trap to survey *C. aeneus* activity across Tift County, Georgia in southern pea fields. We expected that initial trap counts in the spring would correspond to previously published dates of first *C. aeneus* activity in the southern pea crop in Georgia.

2. Materials and methods

2.1. Trap comparison

A new trap design was tested for early detection of *C. aeneus* in 2011 and 2013 at the University of Georgia Lang/Rigdon Farm located in Tifton, Georgia (Latitude 31.515864, Longitude −83.545132). The new *C. aeneus* trap design was modeled after the Tedders trap (Tedders and Wood 1994) (Fig. 1). The modification of the Tedders trap consisted of a standard Leggett style (Leggett and Cross 1971) boll weevil trap (Great Lakes IPM, Vestaburg, MI) baited with extended release 25 mg Grandlure baits (baits provided by the Boll Weevil Eradication Foundation of Georgia Inc., Quitman, GA; Grandlure EPA Pesticide Chemical Code 112404; (E)-(3,3- Dimethylcyclohexylidene) acetaldehyde; (Z)-(3,3-Dimethylcyclohexylidene)acetaldehyde). This trap was attached to the top of two, perpendicular, light yellow, 1-cm thick plastic corrugated triangular sheet fins (CorrugatedPlastics.Net, Hillsborough, NJ) measuring 122 cm × 122 cm × 55 cm base with a 1.5 m × 0.95 cm diameter iron rebar staked into the ground and tied with zip ties to withstand strong wind. In the trap comparison study, a standard Grandlure-baited boll weevil trap was compared with the modified Tedders trap described above and a non-baited 7.6 cm × 12.7 cm yellow sticky card (Great Lakes IPM, Vestaburg, MI) attached to the top of a 30 cm wooden stake. The non-baited sticky trap was similar to what was used in previous pepper weevil trapping studies (Riley and Schuster 1993). Grandlure baits were renewed monthly in the boll weevil and modified Tedders trap.

In 2011, the traps were adjacent to a newly planted southern pea field with 7.6 m between traps. The traps were arranged in a randomized complete block design for two experiments that consisted of 4 replicates each (8 total replications). In 2011, traps were placed in the field on 14 June. *C. aeneus* adults were counted from 17 June to 9 September at an average interval of 2.4 days (range 1–6 days). In 2013, the same trap designs used in 2011 were compared in a randomized complete block design containing 8 replicates located adjacent to a newly planted southern pea field. However, in 2013 we only used 2 m between traps and 60 m between replicates. This was to accommodate field operations. In 2013, traps were placed in

Fig. 1. Modified Tedders trap for *C. aeneus* adults baited with Grandlure (left) and close-up of captured *C. aeneus* adults (right).

the field on 5 March, and curculio adults were counted weekly from 8 March to 13 December. All 8 replicates of adult counts were analyzed together for each year with PROC GLM (SAS, 2010) with trap design as the fixed effect by the date of first detection in the spring, early August and early September when the early and late fall plantings occur. The effect of trap was also assessed for the seasonal average trap count.

2.2. Seasonal sampling

For the cowpea seasonal count study, modified Tedders traps were placed at five separate field locations in Tift County, Georgia from 2012 through 2013. Traps were placed along southern pea field edges, if available at the time of location selection, or next to fallow fields where southern peas were to be planted that year. Grandlure baits were changed at the beginning of each month for all traps in both years. The traps were labeled with a location specific enumeration system. Each location had several replications within the location depending on the southern pea field size. The trap locations in 2012 included the University of Georgia (UGA) Lang/Rigdon Farm off Rigdon-Aultman Road in Tifton Georgia had a total of 13 modified Tedders traps. Five of the traps were (F1 to F5) located along a fence row on the east side of the farm. Four more traps (R1 to R4) were placed on the Lang/Rigdon Farm in the field near the reservoir on Carpenter Road. Four traps (L1 to L4) were added from April to the end of June near the entrance of the farm to increase trap count resolution during this period.

A second location at the UGA Horticulture Farm off Rainwater Road included two traps (S1 and S2). Non-UGA property locations consisted of traps placed on local private farms. At the Robertson Farm located off of Whiddon Mill Road in Tifton, three traps (T1–T3) were set along the field near Whiddon Mill Road. In Turner County, two traps were placed in homeowner gardens (T4–T5). Also, five traps (H1, B1, G1, L1, and P1) were placed in various fields of Griffin Farms close to the county line of Tift and Turner Counties.

For the second year of sampling, most trap locations from 2012 were kept continuous in addition to a few newly located traps. At Lang/Rigdon Farm, R1 to R4 traps were kept continuous and four new traps (R5 to R8) were placed closer to the farm entrance. The Robertson farm traps (T1–T3), and the Horticulture Farm traps (S1 and S2) were continuous. For 2013, the R1–R8 traps were included in both the seasonal sampling data and the comparison study.

Traps for the seasonal count study were checked weekly for *C. aeneus*. In 2012, the sampling dates were approximately twice weekly from 10 January to 3 December at the Lang/Rigdon farm site; trap counts were summed by week before analysis. Additional Lang traps beside Carpenter Road were added on 4 April and ran until 29 June. Additional traps on the Horticulture Farm were added on 13 April and ran until 11 June. Lang Reservoir Traps, and Robertson/Turner traps were added on 15 June and Griffin Farm traps were added on 13 July. Lang Fence traps ran from 20 September to 3 December. Griffin Farm traps ended on 1 October to accommodate field harrowing and commercial operations. In general, the additional field sites were added in 2012 for greater resolution of trap counts during periods of increased *C. aeneus* activity. The 2013 sample dates were approximately weekly from 4 January to 13 December. R5–R8 sticky traps and R1–R4 sticky traps and boll weevil traps were added on 4 March and all were continued on through 13 December. In 2013, approximately 0.4 ha of fall planted southern peas was left in the field at the Lang/Rigdon Farm to allow for late season *C. aeneus* reproduction. For the 2012–2013 survey, a total of 1,710 trap collections were made.

The data from the trap comparison study were analyzed with PROC GLM (SAS, 2010) separating trap means as treatment effects over time and averaged over all sample dates using LSD (P = 0.05) tests only if treatment effects were significant ($P < 0.05$). Trap comparison data for 2011 was plotted as means and standard error values by sample date. The seasonal sampling trap data were averaged over all individual traps by date by year (2012 and 2013), then plotted as an area graph with 2012 in the foreground. Cowpea curculio generation periods of approximately one month (range of 23–53 days with an average of 30.8 days under cowpea growing season field conditions (Arant 1938), were overlaid onto the graph to approximate the generations. Because two distinct late spring peaks of adults on traps were detected in 2013, this was interpreted as either the earliest generation of *C. aeneus* or two distinct invasion events of *C. aeneus* adults from overwintering sites into the southern pea trap site.

In addition, temperature degree days for the spring of 2012 and 2013 were calculated from 1 January to the date of first adult capture on the trap using a minimum development temperature of 6.1°C based on predicted emergence patterns of the boll weevil (Stone et al. 1990). The raw data were collected from the Tift County sites on the Georgia Automated Environmental Monitoring Network at http://www.griffin.uga.edu/aemn/. This value was simply calculated as (Tmax + Tmin)/2 − Tbase, where Tmax is maximum daily temperature, Tmin is the minimum daily temperature, and Tbase is the base temperature for the organism, in this case 6.1°C. This value was used simply to show that lower degree day accumulation was consistent with later emergence, similar to the effect of temperature on emergence of boll weevil adults from overwintering sites (Stone et al. 1990). The mean ± std degree days from January 1 to the first date of *C. aeneus* emergence was estimated.

3. Results

3.1. Trap comparison

The sticky trap was not effective as no *C. aeneus* adults were caught. The modified Tedders trap (Fig. 1) was more effective at capturing adults than the boll weevil trap in both years (Table 1),statistically greater in 2011 than the boll weevil trap on 23 June ($F = 9.3$, df = 2, 14, $P = 0.003$), 24 June ($F = 10.1$, df = 2, 14, $P = 0.002$), 27 June ($F = 8.6$, df = 2, 14, $P = 0.004$), 28 June ($F = 6.5$, df = 2, 14, $P = 0.010$) and July and later on 24 August ($F = 21.3$, df = 2, 14, $P < 0.001$), 2 September ($F = 18.4$, df = 2, 14, $P < 0.001$), 6 September ($F = 20.8$, df = 2, 14, $P < 0.001$) and 9 September ($F = 23.1$, df = 2, 14, $P < 0.001$) (Fig 2).

The modified Tedders trap captured adults approximately one month earlier than the boll weevil trap (Fig. 2). The much greater proportion of adults visiting the modified Tedders trap in 2013 than

Table 1
Comparison of trapping methods for *C. aeneus* adults at the Coastal Plain Experiment Station at Tifton, GA, with traps ($n = 8$) spatially separated from each other from June to September in 2011 and traps spatially clustered together from March to December in 2013.

Treatment	First detection	Early August detection	Mid-September detection	*n*	Overall trap average
2011					
Modified tedders trap	0.5 a	2.4 a	13.3 a	296	3.7 a
Boll weevil trap	0.0 b	1.9 ab	2.9 b	296	1.9 b
Sticky trap	0.0 b	0.0 b	0.0 b	296	0.0 c
2013					
Modified tedders trap	0.13 a	2.75 a	4.0 a	360	0.53 a
Boll weevil trap	0.00 a	0.25 a	0.0 b	360	0.01 b
Sticky trap	0.00 a	0.00 a	0.0 b	360	0.00 b

Means followed by the same letter within columns within years are not significantly different (LSD, $P < 0.05$).

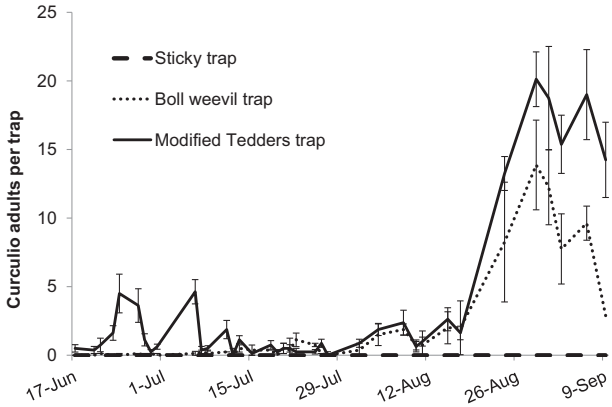

Fig. 2. Comparison of trapping methods over sampling dates based on average individual trap means and corresponding standard error bars ($n = 8$) in 2011.

in 2011 suggested that the close proximity (2 m) between traps favored the modified Tedders trap more than in the more widely spaced (7.6 m) 2011 study (Table 1). Specifically, the modified Tedders trap captured ~2× as many adults as the boll weevil trap in 2011 as compared to 72x as many in 2013 (overall 2011 $F = 35.6$, df = 2, 14, $P < 0.001$ and overall 2013 $F = 6.23$, df = 2, 14, $P < 0.01$ Table 1).

3.2. Seasonal sampling

The first *C. aeneus* adults were caught on 16 April in 2012 and on 30 April in 2013 (Fig. 3). The calculated average degree day for emergence was 954 ± 50, $n = 2$, with the earlier emergence date in 2012 having the greater degree day accumulation value (1004 DD) than 2013 (904 DD). The early invasion of overwintering adults into the trapping area did not occur as a single, compact event, but was staggered over 5 weeks. In 2012, most early adults were captured between mid-April and mid-May; whereas, in 2013, this period extended from early May to late June.

4. Discussion

Overall, the newly modified Tedders trap appeared to be an effective tool for detecting cowpea curculio adults in the field. The lack of captures of adults with this trap in the winter (Fig. 3), suggest that it might also be a way to detect when fall movement to overwintering sites has ended and the first adult movement in the spring into southern pea fields has begun. The detection of adults on traps as an estimate of spring emergence from overwintering sites was consistent with the earliest dates of adult detection on legume crops in the spring, 22 April to 8 May, reported by Arant (1938) and peak occurrence of cowpea curculio on narrow leaf vetch from 30 April to 10 May, a spring time host plant reported by Sudbrink et al. (1998). The new trap design seems to be more effective than boll weevil traps, but it is still unknown how well the trap counts relate to actual *C. aeneus* populations in the field. Still, we can speculate based on reported trends. Arant (1938) reported that only 2–5% of the overwintering *C. aeneus* adult population survived through May. The spring population of overwintering adults should be near the end of their life span; thus, we would expect a decline in the adult numbers in the landscape after their initial reproductive activity in the crop. This was apparent from late-May to late-June in 2012 and from late-June to late-July in 2013 (Fig. 3). Declining pheromone levels could also contribute to the intermittent declines in trap counts because the 3-week baits were only changed monthly (Armstrong 2010). The declines in adult numbers on traps in the middle of the maximum estimated generations per year indicated in Fig. 3 were roughly apparent, but clearly the most significant declines in both years occurred from late-June to early-July and from late-October to early-November (Fig. 3). Thus, there more likely are three distinct generations of *C. aeneus* in southern Georgia and the life cycle estimates of Arant (1938) of 23–53 days per generation likely need more refining to account for the temporal distribution of the three reproductive cycles likely represented by the three clusters of adults captured during the year, i.e., May–June, August–September, and November–December. The large cluster of adults from late-July to early-October more than likely represents the largest reproduction period for *C. aeneus* in the Georgia landscape since this occurs in the warmest time of the year when the greatest acreage of southern peas are growing in the field. Notably, there were distinct declines in adults by mid-October in both years. Since 2–3 weeks of the life cycle of *C. aeneus* is underground as last instar larvae and pupae (Arant 1938), this might represent a period when most of the *C. aeneus* present in the cowpea fields are in the soil.

The general collapse in numbers of *C. aeneus* adults from mid-October to early-November is consistent with empirical observations that commercial fields of flowering southern peas experience less damage during this time of year (C.T. Harvey, Fresh Frozen Inc., personal communication). First, we suspect that *C. aeneus* adults are going into reproductive dormancy in the fall of the year similar to the boll weevil (Brazzel and Newsom 1959, Segers et al. 1987). With boll weevil, adults in full reproductive dormancy are less likely to be captured by pheromone traps (Rankin et al. 1994) and consequently total trap capture rates decline in the fall relative to field counts and are lower in the spring relative to field counts until reproduction begins the next year (Rummel et al. 1975). This theory fits the 2012 seasonal sampling reasonably well. The large difference in the December population between 2012 and 2013 complicates this theory, but could simply be due to the availability of plentiful, late, fall-planted cowpea crop in one year (Fig. 2) and not the other, which would have allowed that portion of the *C. aeneus* population not in reproductive dormancy to reproduce an additional generation, similar to winter reproduction of boll weevil detected in the Lower Rio Grande Valley of Texas when reproductive plant hosts were available in the winter (Wolfenbarger et al. 1976). On the other hand, late season pheromone captures of boll weevil are also known to occur and the reasons are not clear (Spurgeon and Suh 2009). For whatever the reason, this late peak in

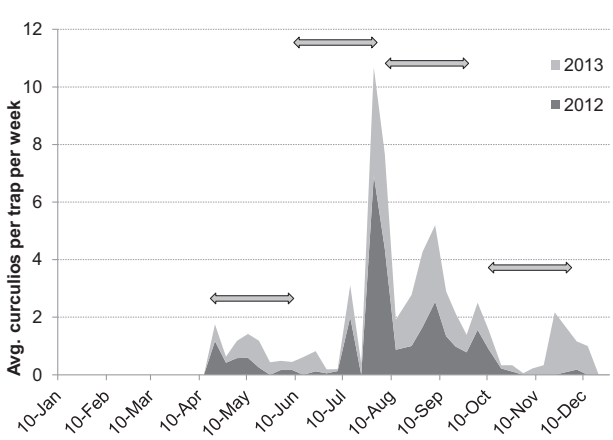

Fig. 3. Seasonal sampling of *C. aeneus* adults based on average trap counts over sampling dates in Tift County, Georgia in 2012 and 2013 with trap means based on inspections of 8 to 15 traps per sample date across three farm sites with an approximate cowpea curculio generation interval (indicated by arrows) based on Arant's (1938) estimate.

C. aeneus captures could potentially be related to the size of the overwintering population of adults and therefore the phenomenon needs to be further investigated if regional management of *C. aeneus* populations is to be attempted.

C. aeneus adults have been reported overwintering in clumps of broomsedge, purple cudweed, heartwing sorrel, moss verbena, and sicklepod (Sudbrink et al. 1998). The May to June movement of *C. aeneus* out of broomsedge reported by Sudbrink et al. (1998) generally coincides with the May to June peaks on traps near cowpea fields reported in this study. An effective early season sampling method for *C. aeneus* is critical for detection and control decisions to possibly prevent the infestation in the southern pea crop. The trap counts from the modified Tedders trap indicate that an effective early season or between season detection of *C. aeneus* is now possible. Similar to the management of boll weevil in cotton, management of *C. aeneus* will likely require more information on whether or not reproductive dormancy exists in this insect, the development of effective control methods for overwintering populations and a regional approach to managing *C. aeneus* in the cowpea cropping systems of the southeastern USA. Since this pest has traditionally been widespread since the early 1900s when cowpea acreage was high, we expect that problems with *C. aeneus* could occur in the region from Texas to Delaware and south to the tip of Florida (Arant 1938). The new modified Tedders trap may be useful in quantifying the current USA distribution of this pest. Additional studies are needed to further determine what trap attractant and trap color would be best for *C. aeneus* as has been done for other weevil pests (Riley and Schuster 1993, Braman et al. 2003).

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